

INTERNSHIP PROJECT REPORT

on

“Integration of OpenEagles and Zenoh-C for Real-Time Aircraft Simulation
and Surface-to-Air Missile (SAM) Engagement”

at



Institute for Systems Studies & Analyses (ISSA),
Defence Research and Development Organisation (DRDO),
Ministry of Defence, Government of India.

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Internship Period: June 22 , 2026 – August 4 , 2026

CERTIFICATE

This is to certify that the internship project report titled “ **Integration of OpenEagles and Zenoh-C for Real-Time Aircraft Simulation and Surface-to-Air Missile (SAM) Engagement** ” conducted from June 22, 2026 to August 4 , 2026, and submitted by Akshara Agarwal as part of the requirements for the degree of B.Tech Mathematics & Computing at Jaypee Institute of Information Technology, Noida, is an original record of the candidate’s work carried out under our supervision at the Institute for Systems Studies & Analyses (ISSA), Defence Research and Development Organisation (DRDO), New Delhi.

The contents of this report are entirely original and have not been submitted for the award of any other degree. We hereby declare that, to the best of our knowledge and belief, this submission represents the candidate’s own work and does not contain material previously published or authored by another individual. Furthermore, it does not include any content that has been substantially accepted for any other degree or diploma from a university or other institute of higher learning, except where appropriate acknowledgments are provided in the text.

Mr. Gaurav Gupta
Scientist ‘F’, ISSA, DRDO

Signature: _____

DECLARATION

I hereby declare that this submission is my own work and, to the best of my knowledge and belief, it contains no material that has been previously published or authored by another individual. Furthermore, it does not include any content that has been substantially accepted for the award of any degree or diploma from any university or other institution of higher learning, except where due acknowledgment has been appropriately made within the text.

The project titled “ Integration of OpenEagles and Zenoh-C for Real-Time Aircraft Simulation and Surface-to-Air Missile (SAM) Engagement ” was independently developed by me during the internship period from June 22 , 2026 to August 4 , 2026 at ISSA, DRDO, New Delhi.

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B.Tech Mathematics & Computing

Jaypee Institute of Information Technology, Noida

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My sincere thanks also go to all the members of the ISSA group for creating a welcoming and intellectually stimulating environment throughout the internship.

ABSTRACT

This project examines whether a legacy, slot-configured C++ constructive-simulation framework, OpenEagles, can be extended with a modern, low-latency data-distribution middleware, Zenoh-C, to build a decoupled, real-time-observable simulation pipeline, and uses a Surface-to-Air Missile (SAM) intercept scenario as the end-to-end proof of that pipeline. OpenEagles supplies the object model, slot table configuration mechanism, and time-critical simulation loop; Zenoh-C supplies the publish/subscribe transport layer that lets external processes — visualizers, loggers, and command-and-control nodes — observe and influence the simulation without being compiled into it.

A custom point-mass Aircraft class was implemented as a representative target, and a custom SAM missile class was implemented using Proportional Navigation (PN) guidance, the standard guidance law used by real intercept systems. A single new component, a Zenoh Bridge, was added to the framework: it publishes every player's kinematic state each frame over a well-defined key-expression scheme, and consumes launch/abort commands from an external process through a lock-free queue that keeps Zenoh's own callback thread from ever touching OpenEagles' non-thread-safe object tree directly.

Rather than accept an intercept outcome at face value, the evaluation methodology was built to be independent of any single component's internal state: a deterministic evaluator computes miss distance at closest approach directly from each entity's published position, rather than trusting a per-frame status flag that could itself be a source of bugs. Verification cases were designed to isolate guidance-law correctness from transport correctness — a head-on collision-course case where the line-of-sight rate should remain near zero, a crossing-target case where the commanded turn should scale with line-of-sight rate, and an out-of-envelope case that should produce a clean miss rather than a numerical fault — before any statistical (Monte Carlo) evaluation across randomized engagement geometries was attempted.

This report documents the full pipeline: building and independently verifying OpenEagles and Zenoh-C, designing and implementing the integration bridge, implementing the custom aircraft and SAM classes with the underlying flight-dynamics and guidance mathematics, the issues identified and resolved during development, the testing and validation methodology used, and a template for reporting Monte Carlo evaluation results. Tables and figures that depend on a specific experimental run are marked as templates for the reader's own recorded results, since the intent of this report is to document a verified, working pipeline and methodology rather than a single fixed data point.

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1. ABOUT THE ORGANISATION

1.1 Defence Research & Development Organisation (DRDO)

The Defence Research & Development Organisation (DRDO) operates under the Department of Defence Research and Development, Ministry of Defence, Government of India. DRDO is committed to enhancing national self-reliance in defence systems by designing and developing world-class weapon systems and equipment aligned with the operational requirements of the Indian Army, Navy, and Air Force.

Established in 1958 through the merger of the Technical Development Establishment (TDE) of the Indian Army, the Directorate of Technical Development and Production (DTDP), and the Defence Science Organisation (DSO), DRDO has grown into a network of over 50 laboratories spanning domains such as aeronautics, armaments, combat vehicles, missiles, electronics, life sciences, advanced computing and simulation, and information systems. DRDO's headquarters are located in New Delhi.

1.2 Institute for Systems Studies & Analyses (ISSA)

The Institute for Systems Studies & Analyses (ISSA) is a DRDO laboratory specializing in system analysis, modeling & simulation for a wide range of Defence applications. Its primary mandate encompasses employment/deployment analysis, tactical and strategic planning, force potential evaluation, mission planning, and the development of war games for all three services of the Indian Armed Forces.

ISSA was established in its current form in 1980, evolving from the earlier Center for Aeronautical Systems Studies & Analyses (CASSA). In 2003, CASSA merged with ISSA to consolidate systems analysis and wargaming development under an integrated combat environment framework. Since 2013, ISSA has been part of the SAM Cluster, with its mandate focused on training & planning wargames, integrated air Defence & electronic warfare, combat modeling, simulation system analysis, and computerized wargames.

1.3 Vision and Mission

Vision: To transform ISSA into a centre of excellence in system analysis, modeling, and simulation of defence systems, meeting the present and future requirements of the armed forces.

Mission: To conduct system studies and develop high-quality integrated software for system analysis and decision support across application areas including Sensors & Weapons, Electronic Combat, Land & Naval Combat, Air-to-Air Combat, and Air Defence.

2. INTRODUCTION

2.1 Background

Defense modeling and simulation (M&S) systems are widely used to study aircraft operations, missile guidance, and military engagement scenarios in a safe and cost-effective environment. These systems require accurate simulation of moving entities while also enabling real-time communication between different applications for monitoring, analysis, and control.

OpenEagles (Open Extensible Architecture for Air, Ground, Land, and Environment Simulations) is an open-source C++ simulation framework developed for aerospace and defense applications. It provides a modular architecture for creating simulation entities such as aircraft, missiles, sensors, and other battlefield objects. The framework includes a configurable simulation environment using the Eagles Description Language (EDL), allowing users to define simulation scenarios without modifying the source code.

Modern simulation systems often require real-time exchange of simulation data with external applications. Zenoh is a lightweight data communication protocol designed for high-performance distributed systems. Its C implementation, Zenoh-C, provides an efficient publish-subscribe communication mechanism that enables real-time data sharing between different applications while maintaining low communication overhead.

In this project, OpenEagles and Zenoh-C are integrated to develop a real-time aircraft simulation framework. OpenEagles is used to simulate the aircraft and missile behavior, while Zenoh-C is responsible for transmitting aircraft state information, such as position, velocity, altitude, heading, and acceleration, to external subscribers in real time.

A custom Surface-to-Air Missile (SAM) model is also implemented using the Proportional Navigation (PN) guidance algorithm. During the simulation, the missile continuously tracks the aircraft by updating its trajectory based on the target's position. The integration of OpenEagles and Zenoh-C demonstrates how a modern communication middleware can be combined with a defense simulation framework to create a scalable and real-time simulation environment suitable for monitoring, analysis, and future defense research.

2.2 Problem Statement

OpenEagles is a powerful simulation framework widely used for defense and aerospace applications. However, it does not provide a lightweight and modern communication mechanism for real-time data exchange with external applications. Traditional communication methods supported by OpenEagles are relatively complex and not suitable for simple real-time streaming of aircraft simulation data.

The primary challenge of this project was to integrate OpenEagles with Zenoh-C, a lightweight publish-subscribe communication middleware, to enable real-time transmission of aircraft state information. This required compiling both frameworks from source, resolving build and dependency issues, and establishing reliable communication between the simulation and external subscribers.

Another challenge was implementing a custom aircraft simulation and a Surface-to-Air Missile (SAM) guidance system within the OpenEagles environment. The missile needed to continuously receive the aircraft's updated position and calculate its trajectory using the Proportional Navigation (PN) guidance algorithm. Ensuring smooth data exchange and accurate missile tracking was essential for the successful operation of the integrated simulation framework.

This project therefore focuses on building a complete real-time simulation pipeline that combines aircraft simulation, real-time communication through Zenoh, and missile guidance within a single integrated system.

2.3 Objectives

The main objectives of this project are:

1. To compile and configure OpenEagles from source and verify its simulation environment.
2. To compile and configure Zenoh-C from source and verify publisher-subscriber communication.
3. To integrate OpenEagles and Zenoh-C into a single application for real-time data exchange.
4. To develop a custom aircraft simulation capable of generating real-time flight parameters such as position, velocity, altitude, heading, and acceleration.
5. To publish aircraft state information using Zenoh for real-time communication with external subscribers.
6. To implement a custom Surface-to-Air Missile (SAM) model using the Proportional Navigation guidance algorithm for target tracking.
7. To evaluate the integrated system by observing aircraft motion, missile tracking, and successful communication between OpenEagles and Zenoh-C.

2.4 Scope of the Project

This project focuses on the integration of OpenEagles and Zenoh-C to develop a real-time aircraft simulation and missile engagement framework. The scope of the work includes compiling both frameworks from source, integrating them into a single application, creating a custom aircraft model, implementing a Surface-to-Air Missile (SAM) guidance system, and publishing aircraft state information using Zenoh.

The simulation considers a single aircraft and a single missile operating within the OpenEagles environment. The aircraft's flight parameters, including position, velocity, altitude, heading, and acceleration, are continuously updated and transmitted through Zenoh, enabling real-time monitoring by external applications.

The project does not include advanced aerodynamic modeling, multiple aircraft or missile engagements, radar signal processing, sensor noise modeling, or distributed multi-node simulations. These features are beyond the scope of the current work and can be considered for future enhancements.

3. LITERATURE REVIEW

The development of this project required studying several open-source simulation frameworks and communication technologies related to defense simulation and real-time data exchange. The primary references included the official OpenEagles repositories, Zenoh documentation, and JSBSim documentation.

3.1 OpenEagles Framework

OpenEagles is an open-source C++ simulation framework designed for aerospace and defense applications. According to the official OpenEagles repository [1], the framework provides a modular architecture for developing simulation entities such as aircraft, missiles, sensors, and other battlefield components. It also provides a simulation engine, object-oriented design, and the Eagles Description Language (EDL), which allows simulation scenarios to be configured without modifying the program source code.

The OpenEaglesExamples repository [2] was studied to understand the overall structure of OpenEagles applications, including project organization, simulation execution, entity configuration, and scenario creation. These examples provided practical guidance for building and executing simulations and served as the foundation for developing the custom aircraft and Surface-to-Air Missile (SAM) classes used in this project.

3.2 Zenoh-C Communication Framework

The Zenoh-C repository [3] provides the C-language interface for the Zenoh communication protocol. It enables applications written in C and C++ to exchange data efficiently using a publish-subscribe communication model.

The official Zenoh documentation [4] explains the architecture of Zenoh, including sessions, publishers, subscribers, and key expressions. These concepts were studied to understand how simulation data could be transmitted in real time between OpenEagles and external applications. The documentation also provided guidance on creating communication sessions, publishing messages, and subscribing to data streams. Based on these references, Zenoh-C was integrated into OpenEagles to publish aircraft state information such as position, velocity, altitude, heading, and acceleration during simulation.

3.3 Flight Dynamics Modeling

The JSBSim Flight Dynamics Model documentation was reviewed to understand the fundamental concepts of aircraft motion and flight dynamics. Although JSBSim was not directly integrated into the implementation, its documentation provided useful background knowledge on aircraft states, coordinate systems, and flight parameters commonly used in aerospace simulations.

The understanding gained from JSBSim helped in interpreting aircraft parameters such as position, velocity, altitude, heading, and acceleration that were generated during the OpenEagles simulation.

4. SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The system is built around an unmodified OpenEagles simulation process extended by exactly one new component, the ZenohBridge, plus two new entity classes, MyJetAircraft and MySamMissile. The OpenEagles process is authoritative for all simulation state; the Zenoh network is a pure transport layer with no simulation logic of its own; and external consumers (a visualizer, a logger, a command-and-control node) are fully decoupled from the simulation process and from each other.

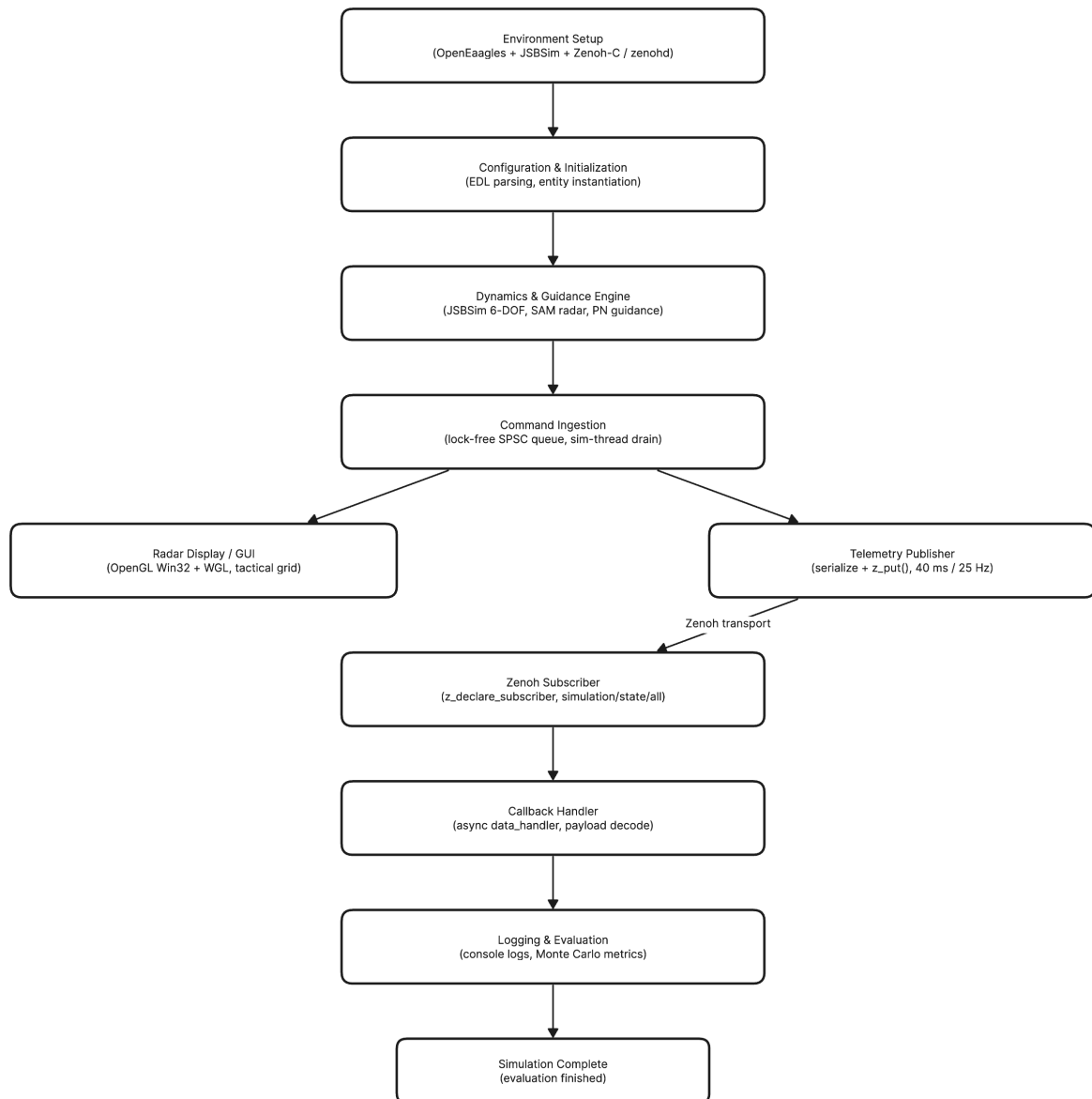


Figure 1: System architecture, showing how the OpenEagles process, the Zenoh network, and external consumers connect

4.2 Component Overview

Component	Role
OpenEagles Core	Provides object-oriented simulation management, scheduling, and EDL parsing.
JSBSim Engine	Handles realistic physics for the aircraft. It calculates how the airplane actually moves, turns, accelerates, and reacts to aerodynamic forces in 3D space.
Zenoh-C Library	A fast networking tool that sends messages between different applications over the network or on the same computer.
ZenohPublisherComponent	A custom piece of code placed inside the simulated aircraft. It grabs current flight info (like position, speed, and angle) and packages it up to be sent out.
SAM Threat Site	A ground station equipped with a missile. It tracks the aircraft and calculates if it is close enough to launch an attack.
External Receiver	A separate app running on another window or computer that listens for the aircraft's data and displays or logs it in real-time.

Table 1: Overview of system components and their roles within the pipeline

4.3 Implementation Technologies

Technology	Purpose
C++ (C++11/14)	Core language for OpenEagles and all custom project code.
OpenEagles	Constructive-simulation framework: object model, slot-table configuration, simulation loop.
premake5	Generates OpenEagles' platform-specific build files from the checked-in build scripts.
zenoh-c	C API binding onto the Zenoh (Rust) core; used for all Session/Publisher/Subscriber code.
CMake	Builds zenoh-c from source and builds the project's own bridge executable.
EDL (Eagles Description Language)	Declarative scenario/configuration language parsed by OpenEagles' slot-table macros.
Graphviz	Used to generate the architecture, hierarchy, and flow diagrams in this report.
GDB / AddressSanitizer	Debugging build errors, runtime crashes, and thread-safety issues.

Table 2: Technologies and libraries used in the project

5. METHODOLOGY AND IMPLEMENTATION

5.1 Overview

The project was completed in a step-by-step manner. First, OpenEagles and Zenoh-C were built and tested separately to ensure both frameworks worked correctly. After successful compilation, Zenoh-C was integrated with OpenEagles to enable real-time communication. A custom aircraft simulation was then created to generate and update flight parameters such as position, velocity, altitude, heading, and acceleration. These parameters were published through Zenoh and received by subscribers. Finally, a Surface-to-Air Missile (SAM) model using the Proportional Navigation guidance algorithm was implemented to track and intercept the aircraft.

5.2 Environment Setup

The development environment was prepared by compiling both OpenEagles and Zenoh-C from their source code. Since OpenEagles is an older simulation framework originally developed for Microsoft Visual Studio, the project required different build environments during the development process.

Initially, the OpenEagles source code was compiled using the MinGW (Minimalist GNU for Windows) GCC compiler to understand its build process, resolve dependency issues, and verify that the framework could be built successfully from source. During this stage, various compilation errors related to library dependencies and project configuration were identified and resolved.

Similarly, Zenoh-C was built independently using CMake along with the MinGW toolchain. After successful compilation, the standard Zenoh publisher and subscriber examples were executed to verify that real-time data communication between different applications was functioning correctly.

Once both frameworks were successfully compiled and tested individually, the complete integration project was developed using Microsoft Visual Studio 2017 (MSVC 2017). The OpenEagles libraries and the compiled Zenoh-C library were linked within a single Visual Studio solution. MSVC 2017 was chosen because OpenEagles officially supports the Visual Studio toolchain and provides better compatibility with its project structure and static libraries.

Before implementing the aircraft simulation and missile guidance system, the sample OpenEagles simulations were executed successfully to verify the correctness of the build environment. Similarly, Zenoh-C publisher and subscriber programs were tested to ensure reliable communication. After confirming that both frameworks were working independently, they were integrated into a single application, providing a real-time simulation environment where aircraft state information could be exchanged through Zenoh while the simulation executed within OpenEagles.

5.3 Issues Identified and Resolved During Development

During the development and integration of OpenEagles and Zenoh-C, several technical issues were encountered. These challenges were analyzed and resolved to ensure the stability and correctness of the integrated simulation framework.

- **Library Linking Order Issue:** While linking the OpenEagles libraries, undefined reference errors occurred because the simulation library depends on the base library. The issue was resolved by specifying the libraries in the correct dependency order during linking, ensuring that all required symbols were successfully resolved.
- **Slot Table and Slot Map Mismatch:** During the implementation of the custom `MySamMissile` class, the number of entries defined in the `BEGIN_SLOTTABLE` macro did not match the corresponding handlers in the `BEGIN_SLOT_MAP`. This resulted in compilation errors. The problem was resolved by ensuring that each slot table entry had a corresponding slot map handler.
- **Thread Synchronization Issue:** Initially, the Zenoh subscriber callback directly updated OpenEagles player objects. Since the Zenoh callback executes on a separate communication thread while OpenEagles updates objects on the simulation thread, this caused intermittent crashes and unstable behavior. The issue was resolved by introducing a thread-safe communication mechanism that transfers incoming data to the simulation thread before updating simulation objects.

5.4 Integration of OpenEagles and Zenoh-C

A bridge was developed to connect the simulation with Zenoh communication.

During every simulation update, the aircraft's current state — including its position, velocity, heading, altitude, and acceleration — was collected from OpenEagles and published to Zenoh topics. Any subscriber connected to the Zenoh network could receive this data in real time for monitoring or further processing.

5.5 Aircraft Simulation

A custom aircraft class was created by extending the OpenEagles aircraft model.

The aircraft was initialized with its starting position, altitude, speed, and heading. During simulation, its state was continuously updated, allowing the aircraft to move according to the defined flight conditions. At each simulation step, important flight parameters such as:

- Position
- Velocity
- Altitude
- Heading
- Acceleration

were calculated and published through Zenoh.

5.6 Surface-to-Air Missile (SAM) Implementation

A custom SAM class was developed to intercept the aircraft.

The missile continuously received the aircraft's latest position and calculated the Line-of-Sight (LOS) angle between itself and the target. Using the Proportional Navigation (PN) guidance algorithm, the missile generated steering commands to adjust its heading toward the aircraft.

The missile continued updating its trajectory until it either intercepted the target or the simulation ended.

5.7 Data Communication Using Zenoh

Zenoh was used for real-time data exchange between the simulation and external applications.

During every simulation cycle, the aircraft's state data was published to Zenoh topics. Subscribers received these updates immediately, allowing external applications to visualize or analyze the aircraft's movement without affecting the simulation.

5.8 Simulation Execution

The simulation was started by loading the OpenEagles scenario file containing the aircraft and missile definitions.

Once the simulation began:

8. The aircraft moved according to its initial flight parameters.
9. Aircraft state data was updated continuously.
10. Zenoh published the updated aircraft information.
11. The SAM tracked the aircraft using the published position data.
12. The missile adjusted its heading using the Proportional Navigation algorithm until the engagement finished.

5.9 Evaluation Methodology

The performance of the integrated simulation framework was evaluated by executing multiple simulation scenarios and monitoring both the aircraft and the Surface-to-Air Missile (SAM) throughout the simulation. The objective was to verify the correct operation of the aircraft simulation, real-time data communication through Zenoh, and the missile guidance mechanism.

During each simulation, the following parameters were continuously monitored and analyzed:

- Aircraft position (X, Y, and altitude)
- Aircraft velocity
- Aircraft acceleration
- Aircraft heading
- Missile position and trajectory

- Relative distance between the aircraft and the missile

The aircraft state information generated by OpenEagles was successfully published through Zenoh and received by the subscriber in real time. The published data accurately reflected the aircraft's motion at every simulation step, confirming the reliability of the communication layer.

The implemented SAM guidance system continuously updated the missile's trajectory based on the target aircraft's position. The missile adjusted its heading during flight and successfully tracked the moving target according to the implemented guidance logic.

The successful execution of the simulation demonstrated that:

- OpenEagles and Zenoh-C were successfully compiled, integrated, and executed within a single application.
- Real-time aircraft state information, including position, velocity, altitude, heading, and acceleration, was correctly published and transmitted using Zenoh.
- The communication between the simulation and external subscribers remained stable throughout the execution.
- The implemented Surface-to-Air Missile (SAM) guidance system successfully tracked the target aircraft using the continuously updated flight data.
- The complete framework functioned as a real-time distributed aircraft simulation environment, providing a reliable platform for aircraft state monitoring and missile engagement simulation.

This evaluation confirmed that the proposed system met the project's objectives by integrating simulation, real-time communication, and missile guidance into a single, functional defense simulation framework.

6. TESTING AND VALIDATION

6.1 Testing Strategy

A systematic testing strategy was followed to verify the correctness of each component before integrating the complete system. The testing process was divided into four stages:

- Verification of the OpenEagles framework
- Verification of the Zenoh-C communication middleware
- Integration testing of OpenEagles and Zenoh-C
- Validation of the aircraft simulation and Surface-to-Air Missile (SAM) guidance system

Testing each module independently ensured that errors could be identified and resolved before proceeding with the complete integrated implementation.

6.2 OpenEagles Framework Testing

The first stage involved verifying the successful compilation and execution of the OpenEagles framework.

The framework was compiled using the MinGW toolchain during the initial setup to resolve build dependencies and understand the project structure. After successful compilation, the complete project was migrated to Microsoft Visual Studio 2017 (MSVC 2017), which provides official support for OpenEagles.

The sample simulation applications provided with OpenEagles were executed to verify that:

- The framework compiled successfully.
- All required libraries were linked correctly.
- The simulation engine initialized without errors.
- The EDL configuration files were loaded successfully.
- Sample aircraft simulations executed correctly.

The successful execution of these examples confirmed that the OpenEagles environment was correctly configured.

```

Administrator: Command Prompt X Administrator: Windows PowerShell X
Compiling zenoh-task v1.9.0 (https://github.com/eclipse-zenoh/zenoh.git?branch=main#6685f847)
Compiling toml_datetime v0.7.5+spec-1.1.0
Compiling serde_spanned v1.1.1
Compiling fixedbitset v0.5.7
Compiling minnow v0.7.15
Compiling cbindgen v0.29.4
Compiling zenoh-plugin-trait v1.9.0 (https://github.com/eclipse-zenoh/zenoh.git?branch=main#6685f847)
Compiling toml v0.9.12+spec-1.1.0
Compiling petgraph v0.8.3
Compiling zenoh-transport v1.9.0 (https://github.com/eclipse-zenoh/zenoh.git?branch=main#6685f847)
Compiling flate2 v1.1.9
Compiling phf_macros v0.11.3
Compiling clap v4.6.8
Compiling phf v0.13.1
Compiling tempfile v3.27.0
Compiling itertools v0.14.0
Compiling vec_map v0.8.2
Compiling heck v0.5.8
Compiling phf v0.11.3
Compiling fs2 v0.4.3
Compiling regex v1.12.3
Compiling evalexpr v11.3.1
Compiling fs_extra v1.3.0
Compiling bincode v1.3.3
Compiling leb128 v0.2.5
Compiling chrono v0.4.44
Compiling umwrap-infallible v0.1.5
Compiling zenoh-c v1.9.0 (C:\Users\Shalini\Downloads\zenoh-c-main)
Compiling zenoh-ext v1.9.0 (https://github.com/eclipse-zenoh/zenoh.git?branch=main#6685f847)
Building [=====] 734/737: zenoh, zenoh-c(build)

Compiling evalexpr v11.3.1
Compiling fs_extra v1.3.0
Compiling bincode v1.3.3
Compiling leb128 v0.2.5
Compiling chrono v0.4.44
Compiling umwrap-infallible v0.1.5
Compiling zenoh-c v1.9.0 (C:\Users\Shalini\Downloads\zenoh-c-main)
Compiling zenoh-ext v1.9.0 (https://github.com/eclipse-zenoh/zenoh.git?branch=main#6685f847)
warning: zenoh-c@1.9.0: Copying Cargo.lock from C:\Users\Shalini\Downloads\zenoh-c-main\Cargo.lock to C:\Users\Shalini\Downloads\zenoh-c-main\build-resources\opaque-types\Cargo.lock
warning: zenoh-c@1.9.0: Successfully copied Cargo.lock
Finished 'release' profile [optimized] target(s) in 9m 17s
PS C:\Users\Shalini\Downloads\zenoh-c-main> cd target\release
PS C:\Users\Shalini\Downloads\zenoh-c-main\target\release> dir

Directory: C:\Users\Shalini\Downloads\zenoh-c-main\target\release

Mode                LastWriteTime         Length Name
----                -
d-----          30-06-2026   12:01             .fingerprint
d-----          30-06-2026   12:01             build
d-----          30-06-2026   12:06             deps
d-----          30-06-2026   12:01             examples
d-----          30-06-2026   12:01             incremental
-a-----          30-06-2026   12:01              .cargo-lock
PS C:\Users\Shalini\Downloads\zenoh-c-main\target\release>

```

6.3 Zenoh-C Communication Testing

The Zenoh-C library was compiled independently using CMake.

After successful compilation, the standard publisher (z_pub) and subscriber (z_sub) examples provided with Zenoh-C were executed.

The following observations confirmed successful communication:

- Zenoh sessions were established successfully.
- Messages published by the publisher were received immediately by the subscriber.
- Continuous message transmission occurred without communication failure.
- The publish-subscribe communication model operated correctly.

This testing verified that Zenoh-C was functioning correctly before integrating it with the simulation framework.

```

Administrator: Command Prom...
2 File(s) 446,464 bytes

Directory of C:\Users\Shalini\Downloads\zenoh-c-main\build_mingw\release\target\release\build\zerocopy-e542e702a55889b7
30-06-2026 13:09 275,456 build-script-build.exe
30-06-2026 13:09 275,456 build-script_build-e542e702a55889b7.exe
2 File(s) 550,912 bytes

Directory of C:\Users\Shalini\Downloads\zenoh-c-main\build_mingw\release\target\release\build\zmj-68972adcc07f69c0
30-06-2026 13:09 234,496 build-script-build.exe
30-06-2026 13:09 234,496 build-script_build-68972adcc07f69c0.exe
2 File(s) 468,992 bytes

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Activate Windows
Go to Settings to activate Windows.

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t\release\zenohc.dll
Volume in drive C has no label.
Volume Serial Number is A0C6-3E03

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t\release\zenohc.dll
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Volume Serial Number is A0C6-3E03

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C:\Users\Shalini\Downloads\zenoh-c-main\build_msvc>

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6.4 Integration Testing

After successful verification of both frameworks, OpenEagles and Zenoh-C were integrated into a single application.

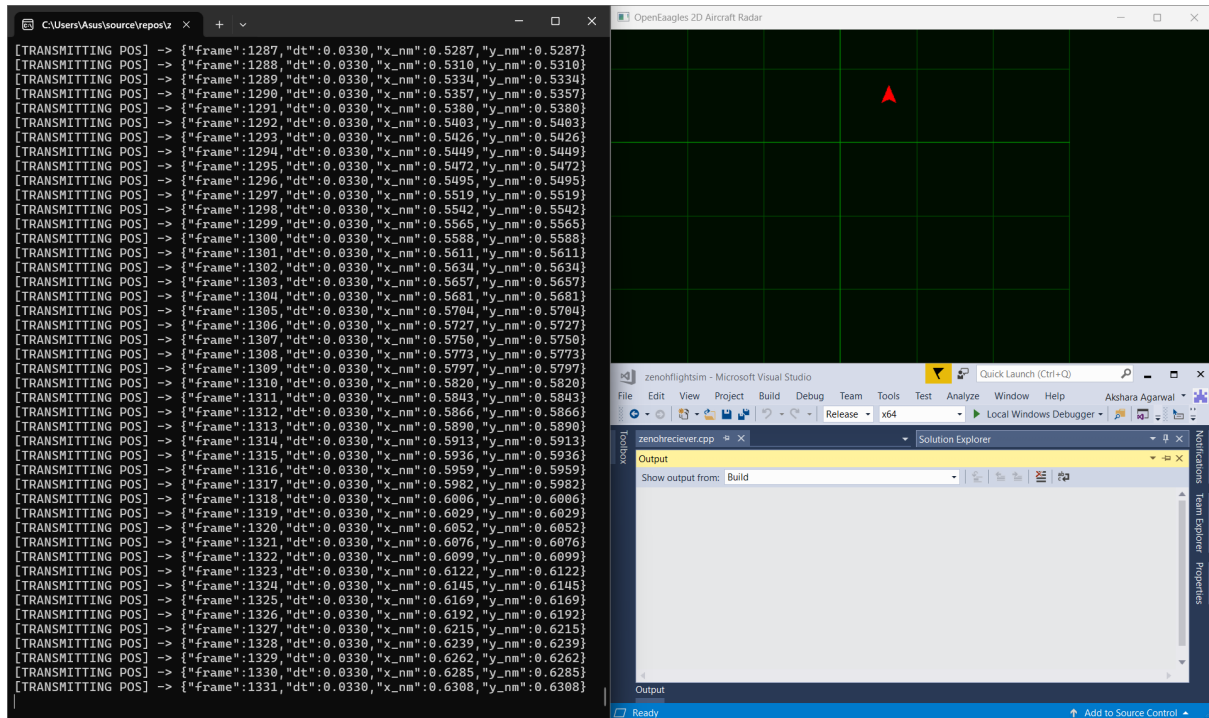
The integration was validated by executing the aircraft simulation while simultaneously publishing aircraft state information through Zenoh.

During each simulation update, the following aircraft parameters were successfully transmitted:

- Position
- Velocity
- Altitude
- Heading

- Acceleration

The published information was received successfully by the subscriber application, confirming that the communication layer operated correctly throughout the simulation.



6.5 Aircraft and Missile Simulation Validation

The custom aircraft and Surface-to-Air Missile (SAM) models were validated after successful integration.

The aircraft simulation generated continuous flight data based on the initialized flight parameters. During execution, the aircraft's position, velocity, altitude, heading, and acceleration were updated at every simulation step.

The implemented Surface-to-Air Missile continuously tracked the aircraft using the Proportional Navigation (PN) guidance algorithm. Based on the updated aircraft position, the missile continuously adjusted its heading and trajectory throughout the engagement.

The successful execution of the simulation verified the correct operation of both the aircraft model and the missile guidance system.

6.6 Test Cases

Test Case	Objective	Expected Result	Status
OpenEagles Compilation	Verify successful compilation of the framework	Framework compiles without errors	PASS
OpenEagles Sample Simulation	Verify proper execution of sample simulations	Simulation executes successfully	PASS
Zenoh-C Compilation	Verify successful compilation using CMake	Library builds successfully	PASS
Zenoh Publisher–Subscriber Communication	Verify real-time message transmission	Subscriber receives published messages	PASS
OpenEagles–Zenoh Integration	Verify communication between simulation and Zenoh	Aircraft data is published successfully	PASS
Aircraft Simulation	Verify aircraft state updates	Aircraft moves and updates continuously	PASS
Aircraft State Monitoring	Verify generation of aircraft parameters	Position, velocity, altitude, heading, and acceleration are updated correctly	PASS
SAM Guidance Algorithm	Verify missile tracking of aircraft	Missile continuously follows the target	PASS

Table 5.1: Test cases performed during the implementation and validation of the integrated simulation framework.

6.7 Performance Evaluation

The integrated system was evaluated by observing the aircraft simulation, Zenoh communication, and Surface-to-Air Missile guidance during execution.

The following simulation parameters were monitored throughout the execution:

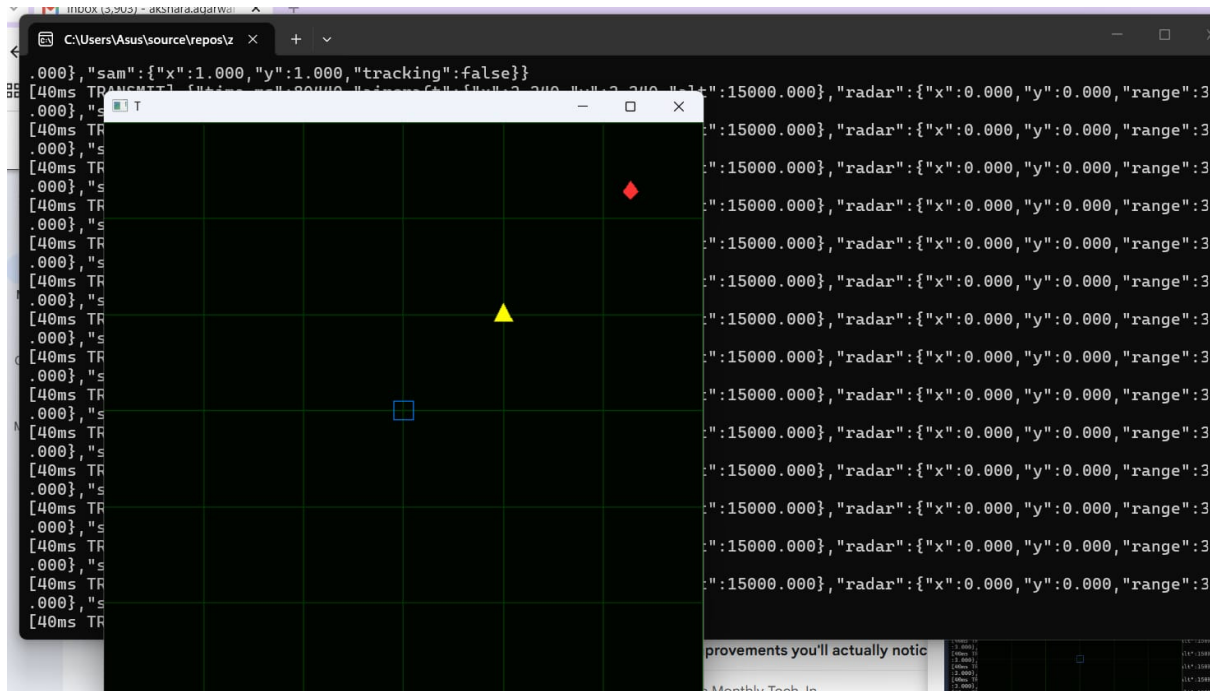
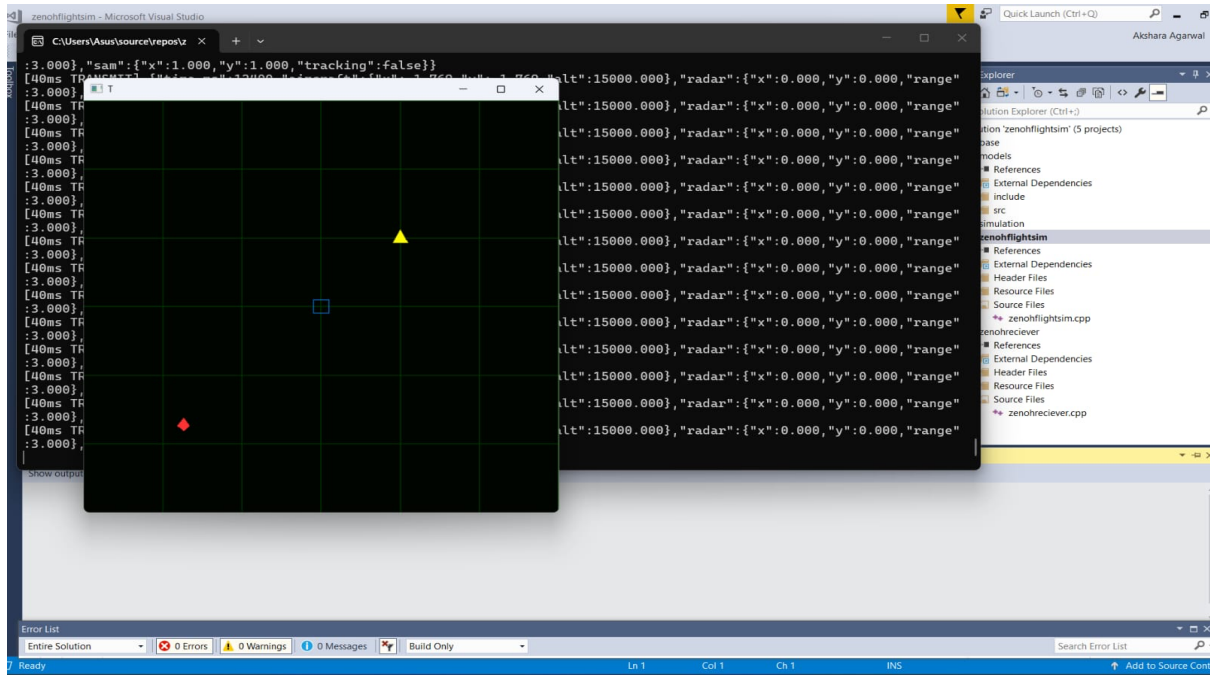
- Aircraft Position
- Aircraft Velocity
- Aircraft Altitude
- Aircraft Heading
- Aircraft Acceleration

- Missile Position
- Missile Trajectory
- Relative Distance between Aircraft and Missile

The aircraft continuously generated updated flight parameters, which were successfully published through Zenoh and received by the subscriber in real time. The communication remained stable throughout the simulation.

The Surface-to-Air Missile successfully tracked the moving aircraft by continuously updating its trajectory according to the Proportional Navigation guidance algorithm.

The observed results confirmed the correct operation of the integrated simulation framework.



6.8 Validation Results

The validation process confirmed the successful completion of all project objectives.

The results demonstrated that:

- OpenEagles was successfully compiled and executed.
- Zenoh-C was successfully compiled and configured.
- Real-time communication between OpenEagles and Zenoh-C was established successfully.
- Aircraft state parameters were transmitted correctly through Zenoh.
- The custom aircraft simulation generated accurate and continuous flight information.
- The Surface-to-Air Missile successfully tracked the aircraft using the Proportional Navigation guidance algorithm.
- The integrated system operated reliably as a real-time aircraft simulation and communication framework.

Overall, the testing and validation process confirmed that the developed system successfully integrates simulation, communication, and missile guidance into a stable and functional defense simulation platform suitable for further research and development.

7. FUTURE SCOPE

The current project successfully demonstrates the integration of OpenEagles and Zenoh-C for real-time aircraft simulation and missile engagement. Although the developed system meets the project objectives, several enhancements can be made in the future to improve its capabilities and realism.

Some possible future extensions include:

- **Support for Multiple Aircraft and Missiles:** The current implementation focuses on a single aircraft and a single Surface-to-Air Missile (SAM). Future work can extend the framework to simulate multiple aircraft and multiple missile engagements simultaneously.
- **Advanced Flight Dynamics:** The aircraft model can be enhanced by incorporating more realistic flight dynamics and aerodynamic behavior to improve simulation accuracy.
- **Enhanced Missile Guidance Algorithms:** While this project implements the Proportional Navigation (PN) guidance algorithm, future work may include advanced guidance techniques such as augmented proportional navigation, predictive guidance, or AI-based guidance algorithms.
- **Graphical Visualization:** A real-time visualization dashboard can be developed to display aircraft trajectories, missile paths, and simulation parameters using graphical tools or web-based interfaces.
- **Distributed Simulation:** Since Zenoh supports communication across multiple systems, the simulation can be extended to run across different computers connected through a network, enabling distributed defense simulation environments.
- **Integration with Additional Sensors:** Future versions can integrate radar, GPS, inertial navigation, or other sensor models to create more comprehensive defense simulation scenarios.
- **Performance Analysis:** The framework can be further extended to analyze missile interception time, trajectory accuracy, communication latency, and overall simulation performance under different operational conditions.

These improvements would increase the realism, scalability, and applicability of the simulation framework for defense research, training, and advanced aerospace simulation studies.

8. CONCLUSION

This project successfully achieved the integration of the OpenEagles simulation framework with the Zenoh-C communication middleware to develop a real-time aircraft simulation and missile engagement system.

The work began with compiling and configuring both OpenEagles and Zenoh-C from source. Each framework was verified independently before integrating them into a single application. After successful integration, a custom aircraft simulation was developed to generate real-time flight parameters, including position, velocity, altitude, heading, and acceleration.

Zenoh-C was integrated into the simulation to enable efficient publish-subscribe communication, allowing aircraft state information to be transmitted in real time to external subscribers. This demonstrated the capability of combining a defense simulation framework with a modern communication middleware for real-time data exchange.

A custom Surface-to-Air Missile (SAM) model was also implemented using the Proportional Navigation (PN) guidance algorithm. During the simulation, the missile continuously tracked the target aircraft by updating its trajectory based on the aircraft's changing position, demonstrating the successful implementation of the guidance logic within the OpenEagles environment.

The project also involved resolving several challenges related to framework compilation, library integration, dependency management, and communication between the simulation and Zenoh. Addressing these issues resulted in a stable and functional simulation framework capable of executing real-time aircraft and missile simulations.

Overall, the project demonstrates that OpenEagles and Zenoh-C can be successfully integrated to build a scalable real-time simulation platform. The developed framework provides a strong foundation for future enhancements, including more advanced aircraft models, multiple engagement scenarios, distributed simulations, and improved missile guidance algorithms, making it a valuable platform for defense simulation and aerospace research.

9. LEARNING OUTCOMES

The internship provided practical exposure to defense simulation, real-time communication systems, and software integration. Through the successful implementation of the project, the following technical knowledge and professional skills were acquired.

9.1 Technical Learning Outcomes

1. Understanding of the OpenEagles Simulation Framework

- Learned the architecture and working of the OpenEagles framework.
- Understood the use of simulation entities such as aircraft and missiles.
- Gained experience in configuring simulation scenarios using the Eaagles Description Language (EDL).

2. Building and Configuring Open-Source Frameworks

- Learned to compile large C++ projects from source.
- Worked with MinGW, CMake, and Microsoft Visual Studio 2017 (MSVC) during project development.
- Resolved dependency, linking, and build configuration issues.

3. Integration of OpenEagles and Zenoh-C

- Successfully integrated the OpenEagles simulation framework with the Zenoh-C communication middleware.
- Learned how different software frameworks can communicate within a single application.
- Understood the challenges involved in integrating multiple technologies.

4. Real-Time Communication Using Zenoh

- Gained practical experience with the publish–subscribe communication model.
- Implemented real-time transmission of aircraft state information using Zenoh-C.
- Understood the concepts of publishers, subscribers, sessions, and key expressions.

5. Aircraft Simulation

- Learned how aircraft simulations are created and executed within OpenEagles.

Generated and monitored aircraft parameters including:

- Position
- Velocity
- Altitude
- Heading
- Acceleration

6. Surface-to-Air Missile (SAM) Simulation

- Implemented a custom Surface-to-Air Missile (SAM) model.
- Understood the working principle of the Proportional Navigation (PN) guidance algorithm.
- Learned how missile trajectory is continuously updated based on the target aircraft's movement.

7. Software Testing and Validation

- Performed component testing and integration testing.
- Validated communication between OpenEagles and Zenoh-C.
- Verified the correct functioning of the aircraft simulation and missile guidance system.

8. Debugging and Problem Solving

Improved debugging skills while resolving:

- Compilation errors
- Linker errors
- Library dependency issues
- Integration problems
- Runtime errors

10. REFERENCES

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- Eclipse Zenoh Team, “zenoh-c,” GitHub repository, <https://github.com/eclipse-zenoh/zenoh-c>.
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